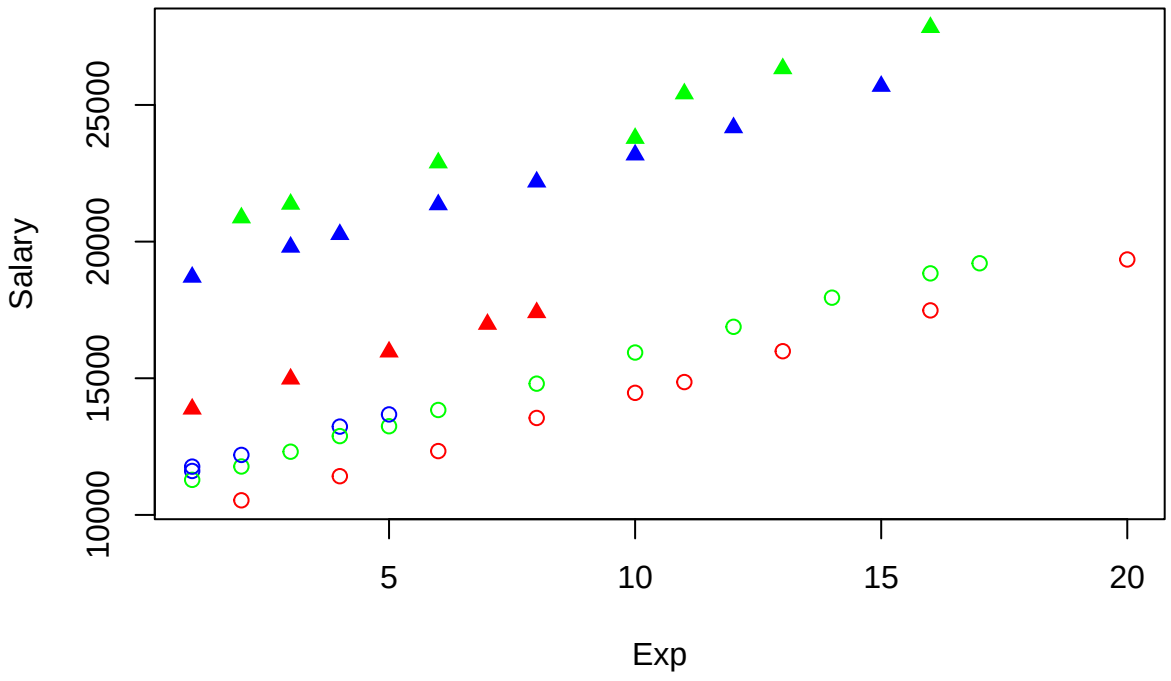
Q.3 a) STAT 4 C03

```
# salary data loading
df = read.csv("Salary.csv")
head(df)
```

```
##   Salary Exp Educ Manage
## 1  13876   1    1      1
## 2  11608   1    3      0
## 3  18701   1    3      1
## 4  11283   1    2      0
## 5  11767   1    3      0
## 6  20872   2    2      1
```

```
cols = c("red", "green", "blue")
syms = c(1, 17)
plot(Salary~Exp, data=df, col=cols[Educ], pch=syms[Manage + 1])
```



Q.3 b)

```
model.b = lm(Salary~Exp * Educ * Manage, data=df)
summary((model.b))
```

```
##
## Call:
## lm(formula = Salary ~ Exp * Educ * Manage, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2678.6  -542.8   -42.8   209.8  3324.2
```

```
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      9087.22    1225.56   7.415  6.8e-09 ***
## Exp               425.87     128.29   3.319  0.00200 **
## Educ              695.48     549.76   1.265  0.21356
## Manage            685.65    1928.96   0.355  0.72422
## Exp:Educ           49.75      71.08   0.700  0.48828
## Exp:Manage        675.48     258.79   2.610  0.01288 *
## Educ:Manage      2519.72     856.71   2.941  0.00554 **
## Exp:Educ:Manage  -264.28     118.13  -2.237  0.03121 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1222 on 38 degrees of freedom
## Multiple R-squared:  0.9434, Adjusted R-squared:  0.9329
## F-statistic:  90.4 on 7 and 38 DF,  p-value: < 2.2e-16
```

The equations of the line has this form: Y = a + bX where a and b takes the following values (Educ = 1, Manage = 0) a = Intercept + Educ b = Exp + Exp:Educ

(Educ = 1, Manage = 1) a = Intercept + Educ + Manage + Educ:Manage b = Exp + Exp:Educ + Exp:Manage + Exp:Educ:Manage

(Educ = 2, Manage = 0) a = Intercept + 2Educ b = Exp + 2Exp:Educ

(Educ = 2, Manage = 1) a = Intercept + 2Educ + Manage + 2Educ:Manage b = Exp + 2Exp:Educ + Exp:Manage + 2Exp:Educ:Manage

(Educ = 3, Manage = 0) a = Intercept + 3Educ b = Exp + 3Exp:Educ

(Educ = 3, Manage = 1) a = Intercept + 3Educ + Manage + 3Educ:Manage b = Exp + 3Exp:Educ + Exp:Manage + 3Exp:Educ:Manage

```r
b_Intercept = coef(model.b)["(Intercept)"]
b_Exp = coef(model.b)["Exp"]
b_Educ = coef(model.b)["Educ"]
b_Manage = coef(model.b)["Manage"]
b_Exp.Educ = coef(model.b)["Exp:Educ"]
b_Exp.Manage = coef(model.b)["Exp:Manage"]
b_Educ.Manage = coef(model.b)["Educ:Manage"]
b_All = coef(model.b)["Exp:Educ:Manage"]

plot(Salary ~ Exp, data = df, col = cols[Educ], pch = syms[Manage + 1])

abline(a = b_Intercept + b_Educ,
       b = b_Exp + b_Exp.Educ,
       col = "red", lwd = 2)        # Educ = 1, Manage = 0

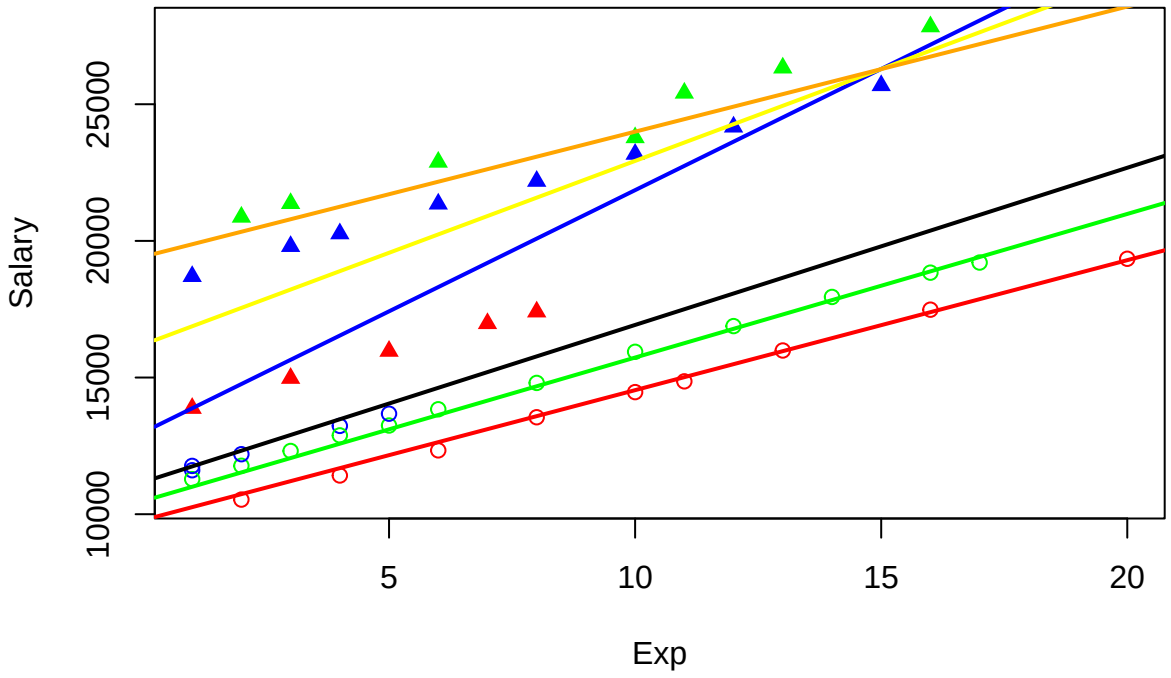
abline(a = b_Intercept + b_Educ + b_Manage + b_Educ.Manage,
       b = b_Exp + b_Exp.Educ + b_Exp.Manage + b_All,
       col = "blue", lwd = 2)       # Educ = 1, Manage = 1

abline(a = b_Intercept + 2*b_Educ,
       b = b_Exp + 2*b_Exp.Educ,
       col = "green", lwd = 2)      # Educ = 2, Manage = 0
```

```r
abline(a = b_Intercept + 2*b_Educ + b_Manage + 2*b_Educ.Manage,
       b = b_Exp + 2*b_Exp.Educ + b_Exp.Manage + 2*b_All,
       col = "yellow", lwd = 2)      # Educ = 2, Manage = 1

abline(a = b_Intercept + 3*b_Educ,
       b = b_Exp + 3*b_Exp.Educ,
       col = "black", lwd = 2)       # Educ = 3, Manage = 0

abline(a = b_Intercept + 3*b_Educ + b_Manage + 3*b_Educ.Manage,
       b = b_Exp + 3*b_Exp.Educ + b_Exp.Manage + 3*b_All,
       col = "orange", lwd = 2)      # Educ = 3, Manage = 1
```



Q.3

c)

```r
model.r = lm(Salary~Exp+Educ*Manage, data=df)
summary(model.r)
```

```
##
## Call:
## lm(formula = Salary ~ Exp + Educ * Manage, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1861.6  -772.9  -131.8   328.5  2869.5
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8007.65     895.54   8.942 3.52e-11 ***
## Exp           547.14      40.07  13.655  < 2e-16 ***
## Educ         1109.25     377.17   2.941  0.00536 **
## Manage       4861.06    1144.01   4.249  0.00012 ***
```

```
## Educ:Manage   910.49     536.01   1.699  0.09697 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1284 on 41 degrees of freedom
## Multiple R-squared:  0.9324, Adjusted R-squared:  0.9259
## F-statistic: 141.5 on 4 and 41 DF,  p-value: < 2.2e-16
```

The six lines are given by the following slope and intercept

(Educ = 1, Manage = 0) a = Intercept + Educ b = Exp

(Educ = 1, Manage = 1) a = Intercept + Educ + Manage + Educ:Manage b = Exp

(Educ = 2, Manage = 0) a = Intercept + 2 · Educ b = Exp

(Educ = 2, Manage = 1) a = Intercept + 2 · Educ + Manage + 2 · Educ:Manage b = Exp

(Educ = 3, Manage = 0) a = Intercept + 3 · Educ b = Exp

(Educ = 3, Manage = 1) a = Intercept + 3 · Educ + Manage + 3 · Educ:Manage b = Exp

Q.3 d)

```
anova(model.r, model.b)
```

```
## Analysis of Variance Table
##
## Model 1: Salary ~ Exp + Educ * Manage
## Model 2: Salary ~ Exp * Educ * Manage
##   Res.Df      RSS Df Sum of Sq      F  Pr(>F)
## 1     41 67624411
## 2     38 56710254  3  10914157 2.4378 0.07945 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The null hypotesis H0 assumes that the following coefficient are null: Exp:Educ = Exp:Manage = Exp:Educ:Manage = 0 We have a p-value of $0.04 < 0.05$ conclusion: H0 is rejected and The true relation doesn't have 6 parallel lines.